Sprint 1: User Input

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Overview

My Sprint 1 task was user input: detecting what the player does with the mouse so they can swap tiles. I added an input handler that lets the player select and swap tiles by clicking or dragging, and I wrote tests for it and for the existing grid code.

How it works

- Left click a tile to select it. A gold outline shows which tile is selected.
- Click a tile next to it to swap the two. Clicking a tile that isn't next to it moves the selection there instead.
- Click the same tile again or click outside the grid to deselect.
- You can also drag a tile onto a neighbor to swap them.
- Right click, middle click, the scroll wheel, and the keyboard don't do anything.

Files changed

- `code/InputHandler.py` (new): handles the mouse events, selection, swapping, and the highlight.
- `code/Grid/Grid.py`: added `ScreenToTile`, which turns a mouse position into a grid position. I moved the layout math from `Render` into `GetLayout` so both use the same numbers, which keeps clicks lined up after the window is resized.
- `code/GameplayManager.py`: creates the input handler and passes it events.

There are two things other parts of the game can use. `onSwap` gets called after every swap, so match detection can check the board or undo a swap that makes no match. Setting `enabled` to False turns input off, for example while tiles are falling.

Menu clicking isn't done yet since there aren't any menus.

Running the tests

From the main repo folder:

```
pip install pygame pytest
python -m pytest
```

Right now this gives 31 passed and 7 xfailed. The xfailed tests are the bugs listed below. They're marked as expected failures so they don't break the test run. When one of the bugs gets fixed, its test will start failing, which means the xfail marker above that test can be removed.

The tests are in the `tests` folder. `test_input.py` covers the mouse input using fake mouse events, so no window opens. `test_board.py` covers the existing grid, swapping, matching, and refill code.

Bugs found

1. `Grid.SwapTiles` returns True but doesn't actually swap anything, and two variable names are misspelled (`postion`, `posiiton`). My input code uses `TileSwap.ValidCheck` instead.
2. `clearMatch` crashes when there is a match because it treats each match as a single position.
3. A row of 4 or more gets counted more than once, so the same tiles show up in several matches.
4. Part of an L or T shaped match can be missed.
5. The starting board can already have matches in it.
6. `Vector2` multiplication rounds to whole numbers.
7. Comparing a vector to None crashes instead of returning False.

Still to do

- Merge with Ryan's mouse input changes once we decide which version to use.
- Add menu clicking once there are menus.
- Add tests for anything new from the team meeting.